1 Solutions 1

1. $\boxed{\text{(B) } 176}$ Read that sequence of numbers very carefully. The 8 is repeated, and the 9 and 13 are omitted. Thus, this sequence is the sum of integers from 1 to 19, $\frac{(19)(20)}{2}$, minus 14. This evaluates to $190 - 14 = 176$

2. $\boxed{\text{(B) } 4 - 4i}$ Evaluate the powers of i first, which is cyclic and repeats every four. $i = i^5$, $i^2 = i^6 = -1$, $i^3 = i^7 = -i$, and $i^4 = i^8 = 1$. Thus, the expression simplifies to $i + 2(-1) + 3(-i) + 4(1) + 5(i) + 6(-1) + 7(-i) + 8(1) = (8 - 6 + 4 - 2) + (-7 + 5 - 3 + 1)i = 4 - 4i$

3. $\boxed{\text{(C) } 3}$ Note that $\det(A + B) \neq \det A + \det B$.

$$\begin{pmatrix} 4 & 15 \\ 0 & 6 \end{pmatrix} - \begin{pmatrix} 0 & 8 \\ -3 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ 3 & 6 \end{pmatrix}$$

The determinant of that is equal to $4 \times 6 - 7 \times 3 = 3$

4. $\boxed{\text{(B) } -1024 + 1024i}$ Note that $(1 - i)^2 = 1 - 1 - 2i = 2i$. Therefore, you can simplify this to $(2i)^{10}(1 - i)$. $i^{10} = i^2 = -1$, so the expression is equivalent to $-2^{10}(1 - i) = -1024 + 1024i$.

5. $\boxed{\text{(B) } 8}$ Instead of re-expanding the entire number, note that 8 is a power of 2. You can actually write each base 8 digit in base 2 and put the entire number together; this will equal the number in base 2. Using this principle, $4_8 = 100_2$, $3_8 = 011_2$, $7_8 = 111_2$, and $5_8 = 110_2$. Putting this all together results in $4375_8 = 100011111110_2$, for which the sum of the digits is 8.

6. $\boxed{\text{(D) Hyperbola}}$ A hyperbola is defined to be the locus of points such that, for two points A and B, the difference between the distance to A and the distance to B is constant. The difference between the distances from z to $3 - 4i$ and from z to $\ln 2$ is constant, so the locus is a hyperbola.

7. $\boxed{\text{(B) } 10}$ Note that if $S_n = f(n) = An^2 + Bn + C$, then $A - B + C = f(-1)$. We can ignore the fact that S_n is true only for $n \geq 1$, so using finite differences, we get that $f(-1) = A - B + C = 10$.

8. $\boxed{\text{(C) } (8, 12)}$ The test initially contains $0.2(20) = 4$ Geometry questions and $0.8(20) = 16$ Algebra questions. After adding some $A + C$ questions, the total Algebra questions becomes $16 + A$, the total Geometry stays the same (4), and the total question count becomes $20 + A + C$. Since the Geometry question count stays the same but its percentage is halved, 20 extra questions must have been added. This means $A + C = 20$ and the total question count is 40. Furthermore, $\frac{16+A}{40} = \frac{3}{5}$, so $16 + A = 24$ and $A = 8$. $C = 20 - 8 = 12$, so the ordered pair is $(8, 12)$.

9. $\boxed{\text{(B) } 32}$ Remainder theorem gives us that the remainder is equal to

$$9^{2024} - 81(9^{2022}) + 81 - 54 + 5 \implies 9^{2024} - 9^{2024} + 32 \implies 32$$

10. $\boxed{\text{(D) } 10}$ For a Three-Card Spread, we can initially choose 3 cards for the reading, giving us $\binom{22}{3}$. Then, each of these cards can be in any order and can be either in upright position or reserved. Thus, we get that the number of distinct Three-Card Spreads is

$$\binom{22}{3} \times 3! \times 2^3 = 2^6 \times 3 \times 5 \times 7 \times 11$$

Thus, $a + b + c + d + e = 10$

11. $\boxed{\text{(C) } 242}$ For the sum of all coefficients in that expanded multinomial expression, one would plug in 1 for w, x, y, and z. Using a similar principle, you can find the sum of the coefficients of all the terms that don't have an x term and subtract the total from that. The total sum of the coefficients is $(2(1) + 2(1) - 2(1) - 3(1))^5 = -1$. The sum of coefficients of all the terms without an x term is $(2(1) + 2(0) - 2(1) - 3(1))^5 = (-3)^5 = -243$. The sum we're finding is going to be $(-1) - (-243) = 242$.

12. $\boxed{\text{(A) } -4}$ By Vieta's, we get that $a+b+c=-1$ This implies that

$$b+c=-1-a$$
$$a+c=-1-b$$
$$a+b=-1-c$$

This substituting these values in, we get

$$\frac{b+c}{a^2+a}+\frac{a+c}{b^2+b}+\frac{a+b}{c^2+c}=-\frac{1}{a}-\frac{1}{b}-\frac{1}{c}$$

By Vieta's, we get that this is equal to $-\dfrac{12}{3} \implies -4$

13. $\boxed{\text{(B) } 72}$ Denote the point of tangency between the circle and $\overline{AJ}$ as K. By basic POP, we can see that $AA' = AK$ and $JJ' = JK$ and $AK + JK = AJ$. Additionally, $RA' = RA + AA'$ and $RJ' = RJ + JJ'$. The perimeter of $\triangle RAJ \implies RA+AK+JK+RJ \implies RA+AA'+RJ+JJ' \implies RA'+RJ'$. This computes to $32+40 \implies 72$

14. $\boxed{\text{(A) } 1}$ By Cramer's Rule, we get that

$$x=\frac{\begin{vmatrix}1 & 4 & -6 & 1\\ -2 & 0 & 7 & -9\\ 3 & 9 & -1 & 5\\ -8 & -1 & 1 & -1\end{vmatrix}}{\begin{vmatrix}1 & 4 & -6 & 1\\ -2 & 0 & 7 & -9\\ 3 & 9 & -1 & 5\\ -8 & -1 & 1 & -1\end{vmatrix}}$$

These matrices are the same, so their determinants are the same. Thus, $x = 1$.

15. $\boxed{\text{(C) } 102}$ Taking the reciprocals of the entire inequality statement, you have $\frac{2048}{5} < \frac{x+y}{x} < \frac{512}{1}$. You can turn $\frac{x+y}{x}$ into $1+\frac{y}{x}$ so you can subtract 1 from both sides. This results in $\frac{2043}{5} < \frac{y}{x} < 511$. $\frac{2043}{5} = 408.6$, so the least integer greater than this is 409 and the greatest integer less than 511 is 510. There are $510-409+1 = 102$ numbers between these two inclusive bounds.

16. $\boxed{\text{(B) } 85^\circ}$ Let x and y be the other 2 side lengths. By triangle inequality, we are given that $x+1 > y$ and $y+1 > x$. Since x and y are both integers, these two inequalities are both satisfies if and only if $x = y$. Thus, $\triangle BLS$ is isosceles. This implies that $m\angle S \approx \left(\dfrac{180^\circ - 10^\circ}{2}\right) \approx 85^\circ$

17. $\boxed{\text{(A) } -1}$ $2^{10}+2^5$ can be factored as $2^5(2^5+1) = 32(33)$. Similarly, $K^{10}+K^5$ can be factored as $K^5(K^5+1)$. One solution is pretty trivial, where $K^5 = 32$. Another solution can be found when considering the fact that $(32)(33) = (-33)(-32)$. K is actually allowed to be any real number, so K^5 can equal -33. This makes $K^5+1 = -32$, and the inequality holds. Adding the two solutions for K^5 we found gives $32+(-33) = -1$.

18. $\boxed{\text{(C) } 2\sqrt{2}}$ By the given, $\cos Y = \dfrac{5}{\sqrt{26}}$. Thus, by Law of Cosines, $XZ^2 = XY^2+YZ^2-2\cdot XZ\cdot XY\cdot\cos Y$ Plugging everything in, we get

$$XZ^2 = 104+64-2\cdot 2\sqrt{26}\cdot 8\cdot\frac{5}{\sqrt{26}} \implies XZ^2 = 8 \implies XZ = 2\sqrt{2}$$

$\cos Y = \dfrac{5}{\sqrt{26}} \implies \dfrac{5}{\sqrt{26}} = \dfrac{104+64-X^2}{4\sqrt{26}\cdot 8} \implies 160 = 168 - X^2$
$X^2 = 8 \implies X = 2\sqrt{2}$

19. $\boxed{\text{(D) } 3\sqrt{2}}$ Let the perpendicular from M to RU intersect RU at X. Using Geometric means / similar triangles, we get that $\dfrac{RU}{UM} = \dfrac{UM}{UZ} \implies UM^2 = 3 \implies UM = \sqrt{3}$. Since $MR^2+UR^2 = UR^2$, we can easily find that $MR^2 = 6 \implies MR = \sqrt{6}$. Thus, the area of the rectangle is $\sqrt{3}\times\sqrt{6} = 3\sqrt{2}$

20. $\boxed{\text{(B) } 20}$ Let $y = x^2 + 18x + 30$. Substituting this, we get that

$$x^2 + 18x + 30 = 2\sqrt{(x^2 + 18x + 30) + 15} \implies y = 2\sqrt{y + 15}$$

Solving for y, we get

$$y^2 = 4y + 60 \implies y = 10, -6$$

However, $y = -6$ is extraneous, meaning the only solution is $y = 10$. Plugging this value back in, we get $x^2 + 18x + 30 = 10 \implies x^2 + 18x + 20 = 0$. Finally, by Vietta's, we get that the product of the roots is 20.